

A 3-adic point on the five-row comparison variety

Exact computational certificate; independent review requested

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Proposition 1. *There are four columns $x_1, \dots, x_4 \in \mathbb{Q}_3^5$ satisfying*

$$\frac{1}{5} \sum_{i=1}^5 \prod_{j \in S} x_{j,i} = \begin{cases} 0, & |S| \text{ odd}, \\ 1/(|S| + 1), & |S| \text{ even}, \end{cases} \quad \emptyset \neq S \subseteq \{1, 2, 3, 4\}.$$

In particular, failure of local solubility over $\mathbb{Q}_3$ cannot exclude a rational five-row comparison model.

Proof. Write $x_j = z_j/3$. The desired integral polynomial equations are

$$F_S(z) := \sum_{i=1}^5 \prod_{j \in S} z_{j,i} - b_S = 0, \quad b_S = \begin{cases} 15, & |S| = 2, \\ 81, & |S| = 4, \\ 0, & |S| \text{ odd}. \end{cases}$$

Take the following four initial columns:

$$z_1 = (1, 7, 3, 15, 1), \quad z_2 = (19, 25, 3, 15, 19), \quad z_3 = (2, 15, 7, 11, 19), \quad z_4 = (5, 11, 15, 0, 23).$$

Order the nonempty subsets first by size and then lexicographically. The residual vector is

$$(27, 81, 54, 54, 432, 297, 135, 945, 837, 702, \\ 5562, 2592, 1917, 12933, 38232),$$

so every entry is divisible by 27. Flatten the variables column by column, using indices $0, \dots, 19$. The Jacobian minor using all 15 equations and variable indices

$$0, 1, 2, 3, 5, 6, 7, 8, 10, 12, 15, 16, 17, 18, 19$$

has determinant 13966477728153600, of 3-adic valuation one. These integer calculations are reproduced by `lift_general_q3.py`, and recorded in `general_q3_hensel_certificate.json`.

Fix the other five coordinates. If J denotes this square Jacobian at the displayed point, its inverse has entries in $3^{-1}\mathbb{Z}_3$ by the adjugate formula. On the ball $z + 9\mathbb{Z}_3^{15}$ the map $y \mapsto y - J^{-1}F(y)$ preserves the ball: its displacement at the center lies in $9\mathbb{Z}_3^{15}$, and its derivative has entries in $3\mathbb{Z}_3$. It is therefore a strict contraction on this complete ball. Its fixed point is an exact zero of F , proving the assertion. $\square$

This is a local-solubility certificate only. It asserts neither a rational point nor a monotone real point, and does not address other primes or the realization of a local model by actual dice.